INDIAN INSTITUTE OF TECHNOLOGY ROPAR • CSE DEPARTMENT

AI511: Human-Computer Interaction and Visualization • Autumn Semester 2026

LAB 9: VALIDATE — EVALUATION OF IMPROVED BUILD

Triangulated Validation across Usability Inspection, User-Based Testing, and Accessibility Audits for Dash

EVALUATION MILESTONE	EVALUATION DATE	TEAM MEMBERS	TESTED SYSTEM
Lab 9: Validation Summary	14 October 2026	Harshit (Lead) & Uttam Kumar	Dash v3.0 (Jaipur ↔ Delhi)

1. Executive Summary & Theoretical Grounding

The core mandate of **Lab 9: Validate** is to empirically and systematically verify whether our **Lab 8 improvements** successfully resolved the critical usability friction, heuristic defects, and accessibility barriers uncovered during **Lab 7**. To ensure absolute methodological validity, this evaluation adopts the tripartite evaluation taxonomy established in *Labs 1–3 (Slides 8–10)*:

Evaluation Dimension	Methodological Definition	Instruments Employed in Lab 9	Primary Operational Objective
1. Usability Inspection	Expert-driven inspection of the interactive artifact to detect design flaws early without user interference.	Jakob Nielsen's 10 Usability Heuristics re-audit, Cognitive Walkthrough of 4-step wizard.	Verify that all 10 Lab 7 heuristic violations are eliminated or mitigated to severity ≤ 1.
2. User-Based Evaluation	Empirical observation and measurement of real representative users performing realistic target tasks.	5 completely new external participants (N01–N05), Concurrent Think-Aloud, standardized SUS.	Quantify performance velocity, eliminate decision errors, and benchmark satisfaction gains.
3. Accessibility Evaluation	Systematic auditing of technical barriers to ensure equitable access across diverse sensory abilities.	Google Lighthouse (v11), WAVE Accessibility Evaluator, WebAIM Contrast Checker (WCAG 2.1 AA).	Guarantee compliant color contrast (≥ 4.5:1), complete ARIA labelling, and pure keyboard navigation.

87.50 / 100	3.8 s	6.4 s	10 / 10
MEAN SUS SCORE	TASK 1 COMPLETION MEAN	TASK 2 COMPLETION MEAN	HEURISTIC ISSUES RESOLVED
▲ +13.41 pts (Grade A / Excellent)	▼ -60.8% Faster (vs 9.7s Lab 7)	▼ -51.9% Faster (vs 13.3s Lab 7)	100% Catastrophic/Major Fixed

2. Pre-Set Measurable Targets for Key Tasks (from Lab 5)

In accordance with Lab 9 guidelines, our team established **4 rigorous, measurable quantitative targets** prior to running user validation. These targets directly correspond to the **2 core operational tasks** defined in Lab 5:

- **Task 1 (Journey Discovery & Train Selection):** *"Plan a journey from Jaipur (JP) to Delhi for a target future date, evaluate available train timings and terminal differences using Station Intelligence, and select the best train option."*
- **Task 2 (Intelligent Passenger Allocation & Review):** *"Add 2 passengers (including a senior citizen age 62 requiring lower berth priority and a companion), configure class and berth preferences, review the itemized fare breakdown, and confirm simulated checkout."*

Target ID	Evaluation Dimension	Lab 7 Baseline Metric	Lab 9 Pre-Set Target	Design Rationale & Grounding
Target 1	**Task 1 Efficiency** (Search to Selection Time)	9.7 s mean (Outlier: 39s by P02)	**≤ 4.5 seconds mean** (≥ 50% reduction)	Date chips replace manual typing; "Delhi — All Stations" cluster eliminates terminal guessing.
Target 2	**Task 2 Efficiency** (Passenger Entry & Review)	13.3 s mean (Outlier: 53s by P07)	**≤ 8.0 seconds mean** (≥ 40% reduction)	Group constraint engine auto-allocates lower berths for seniors; inline editing removes modal friction.
Target 3	**Subjective Satisfaction** (Standardized SUS Score)	74.09 / 100 mean (Grade B+, Range 60–87.5)	**≥ 85.0 / 100 mean** (Grade A / Industry Elite)	Progressive disclosure, live availability trust badges, transparent fare breakdown boost user confidence.
Target 4	**Cognitive Error & Hesitation Rate**	72.7% friction rate (8/11 verbalized confusion)	**0 terminal dropouts, ≤ 1 minor hesitation**	Eliminate terminal ambiguity questions, explain Smart Match trade-offs, clarify demo boundary.

3. User Validation (Evaluation Plan: 2 Marks)

3.1 Methodological Justification & User-Based Protocol

Following the HCI principles from Rogers, Sharp & Preece and Lab 3 Slide 9, *User-Based Evaluation* captures authentic cognitive behavior that cannot be modeled through expert inspection alone. We enforced two strict scientific controls:

1. **Absolute Sample Independence:** We recruited **5 completely new participants (N01–N05)** outside our team who had never interacted with Dash in Lab 7. This eliminated *learnability bias* and muscle memory contamination, ensuring true ecological validity.
2. **Concurrent Think-Aloud & Non-Intervention:** Participants verbalized cognitive expectations throughout both tasks while evaluators logged task duration via millisecond precision timers, recording hesitations, backtracking, and verbal confusion.

3.2 Empirical Validation Session Log (5 New Participants N01–N05)

ID	Demographic Profile	T1 Time	T2 Time	Errors	Selected Train	SUS	Think-Aloud Log & Qualitative Verification
N01	Tech-Savvy Student Commuter (IIT Ropar)	2.8 s	4.9 s	0	Vande Bharat Exp (20977)	92.5	*"Clicking the date chip was instantaneous. Station intelligence immediately made it obvious Delhi Cantt arrives fastest."*
N02	Occasional Festival Traveler (Non-Tech)	4.6 s	7.8 s	0	Swarna Jayanti Rajdhani (12957)	85.0	*"I loved that the screen didn't bombard me. Step 1 then Step 2. The green LIVE badge gave me immediate trust in seat numbers."*
N03	First-Time Digital Rail Planner (Prefers Counters)	5.1 s	8.2 s	0	Uttaranchal Exp (19565)	80.0	*"Normally IRCTC confuses me with quotas. Here General was preselected and the price breakdown for 2 people was very clear."*
N04	Family Travel Planner (Senior Accompaniment)	3.4 s	6.1 s	0	Mandore Express (22996)	90.0	*"Entering age 62 automatically prioritized Lower Berth for my father without me having to search through buried menus."*
N05	Frequent Business Commuter (Delhi Corridor)	3.1 s	5.0 s	0	Ajmer Shatabdi (12016)	90.0	*"The Smart Match card actually explained the trade-off between breakfast catering vs 4:30 AM wake-up. That's real intelligence."*

Empirical Summary for New User Cohort (N01–N05):

100% Task Completion Rate across both tasks without evaluator intervention. Mean Task 1 duration dropped to **3.80 seconds**; mean Task 2 duration reached **6.40 seconds**; mean System Usability Scale (SUS) reached an exceptional **87.50 / 100**.

3.3 Rigorous Before-and-After Comparison: Lab 7 vs. Lab 9

Evaluation Metric	Lab 7 Build (Baseline)	Lab 9 Build (Validated)	Observed Delta	Statistical Status
User Cohort Sample	11 Participants (P01–P11)	5 New Participants (N01–N05)	Independent sample	VALID CONTROL
Task 1 Mean Time	9.7 seconds (Max: 39.0s)	3.80 seconds (Max: 5.1s)	**-5.90 s (-60.8%)**	TARGET MET
Task 2 Mean Time	13.3 seconds (Max: 53.0s)	6.40 seconds (Max: 8.2s)	**-6.90 s (-51.9%)**	TARGET MET
Task Completion Rate	100% (11/11)	100% (5/5)	0.0% (Maintained)	OPTIMAL
Mean SUS Score	74.09 / 100 (Grade B+)	87.50 / 100 (Grade A)	**+13.41 points**	TARGET MET
SUS Score Range	60.0 to 87.5 (SD: 7.39)	80.0 to 92.5 (SD: 4.87)	Variance compressed	HIGH CONSISTENCY
Cognitive Friction Rate	72.7% (8/11 hesitations)	0.0% terminal errors	**-72.7% reduction**	TARGET MET

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4. Inspection & Accessibility Evaluation (Evaluation Plan: 2 Marks)

4.1 Status of Lab 7 Heuristic Issues (Nielsen's 10 Usability Heuristics Re-Audit)

In accordance with *Lab 3 Slide 9 (Usability Inspection)*, our evaluators conducted an exhaustive expert re-audit of the improved build against Jakob Nielsen's 10 Usability Heuristics. Every single defect flagged during Lab 7 was evaluated for remediation status:

ID	Component	Heuristic Violated	Lab 7 Sev.	Lab 8/9 Concrete Architectural & Interaction Fix	Lab 9 Status	Post Sev.
H1	Search Date Selector	5. Error prevention	2 (Minor)	Replaced free-text typing with interactive 14-day date chip strip and validated calendar selector. Zero syntax errors.	FIXED	0 (None)
H2	Monolithic Main Layout	8. Aesthetic & minimalist	3 (Major)	Enforced strict Progressive Disclosure: linear 4-step wizard (Journey → Select Train → Passengers → Review). Secondary panels moved to slide-out drawers.	FIXED	0 (None)
H3	Delhi Terminal Ambiguity	2. Match system & real world	3 (Major)	Implemented unified "Delhi — All Stations" cluster with Station Intelligence panel explaining transit benefits of NDLS vs DLI vs DEC.	FIXED	0 (None)
H4	Opaque Smart Match	3. User control & freedom	3 (Major)	Deployed Smart Match 2.0 with explainable pros & trade-offs badges. Users control priority toggles (Fare, Arrival, Comfort, Best Chance).	FIXED	0 (None)
H5	Static Seat Count & Trust	1. Visibility of system status	4 (Catastrophic)	Integrated real-time RailKit API cascade (`railkit@6.0.1`), verified live IRCTC counts, green LIVE badge, sync timestamp, and transparent fallback badges.	FIXED	0 (None)
H6	Aggregate Fare Opacity	1. Visibility of system status	3 (Major)	Itemized base fare per passenger, optional meal selection (₹120/meal), and comprehensive total payable breakdown before confirmation.	FIXED	0 (None)
H7	Rigid Group Booking	7. Flexibility & efficiency	3 (Major)	Added Intelligent Group Constraint Engine: automatically detects senior citizens (age ≥ 60) for lower berth priority and pairs children with adults.	FIXED	0 (None)

ID	Component	Heuristic Violated	Lab 7 Sev.	Lab 8/9 Concrete Architectural & Interaction Fix	Lab 9 Status	Post Sev.
H8	Master-List Preference Flaw	7. Flexibility & efficiency	3 (Major)	Enabled in-flow passenger preference editing with per-leg overrides for connecting trains, solving IRCTC's classic master-list constraint.	FIXED	0 (None)
H9	Absence of Recovery Path	9. Help recognize & recover	3 (Major)	Surfaced proactive Recovery Journey recommendations (Shatabdi/Rajdhani backups) and journey-change alert intent recording without restarting flow.	FIXED	0 (None)
H10	Ambiguous Booking Boundary	10. Help and documentation	4 (Catastrophic)	Clarified prototype boundaries: re-badged checkout as "Simulated Secure Payment Handoff" with persistent Data & Trust transparency modal.	FIXED	0 (None)

4.2 Accessibility Evaluation Results (WCAG 2.1 AA Standards)

Following *Lab 3 Slide 9 (Accessibility Evaluation)*, the build was audited using automated evaluators (Google Lighthouse, WAVE) and manual keyboard/screen-reader testing across four core dimensions:

Audit Dimension	Lab 7 Baseline Finding	Lab 9 Remediation & Audit Results	WCAG 2.1 Status
Color Contrast Ratios (WebAIM Checker)	Secondary caption text (#64748b) exhibited 3.8:1 contrast on dark containers, failing AA threshold.	Adjusted text color tokens to #94a3b8 (dark mode: 5.4:1) and #475569 (light mode: 7.2:1). Primary text achieves 14.2:1.	PASSED AA (100%)
Keyboard Navigation & Tab Order	Controls reachable but focus indicator was faint (1px browser dotted outline).	Implemented high-contrast focus rings (2px solid #6366f1 with 2px offset). Full linear Tab/Shift-Tab order through all 4 wizard steps.	PASSED AA (100%)
Typography & Visual Hierarchy	Tagline metadata rendered at 10px, causing visual fatigue for low-vision readers.	Minimum font size raised to 12px (0.75rem); labels paired with high-clarity SVG iconography. Responsive rem sizing throughout.	PASSED AA (100%)
Screen-Reader & ARIA Labelling	Icon buttons and custom chips lacked explicit `aria-label` attributes.	Added programmatic `aria-label`, `aria-expanded`, `role="radiogroup"`, and semantic form ` ` tags across all interactive controls.	PASSED AA (100%)
Google Lighthouse Accessibility Score	Baseline Score: **82 / 100** (Defects: color contrast, missing button names).	Validated Score: **98 / 100** (Zero critical accessibility violations detected by automated audit engine).	98 / 100 (ELITE)

5. Validation Summary & Project Completion Plan (Evaluation Plan: 2 Marks)

5.1 Measurable Target Achievement Scorecard

Pre-Set Target	Target Criteria	Empirically Observed Result	Evaluation Status	Conclusion
Target 1	Task 1 Mean Time ≤ 4.5s	**3.80 seconds** (Range: 2.8s – 5.1s)	TARGET MET	Exceeded goal by 0.70s (60.8% faster than Lab 7).
Target 2	Task 2 Mean Time ≤ 8.0s	**6.40 seconds** (Range: 4.9s – 8.2s)	TARGET MET	Exceeded goal by 1.60s (51.9% faster than Lab 7).
Target 3	Mean SUS Score ≥ 85.0	**87.50 / 100** (Range: 80.0 – 92.5)	TARGET MET	Elevated product into Grade A "Excellent" tier (+13.41 pts).
Target 4	Zero terminal errors, ≤ 1 hesitation	**0 dropouts, 0 errors, 1 minor query**	TARGET MET	100% completion across all 5 new participants.

5.2 Remaining Project Limitations & Transparent Scope Boundaries

Honest Academic Prototype Boundaries (Defensible for Course Viva):

- **Commercial IRCTC Ticketing Gateway:** Dash simulates the final payment handoff. Direct ticket issuance requires government-licensed IRCTC commercial partner APIs, which are legally restricted to commercial aggregators.
- **Geographic Corridor Focus:** Dash is intentionally optimized as a high-density prototype for the *Jaipur ↔ Delhi corridor*. Expanding nationwide requires multi-node graph routing across thousands of Indian railway stations.
- **Native Voice/AI Assistant:** While conversational natural-language booking ("Book me a morning Shatabdi for tomorrow") is conceptually designed, it is scheduled for integration during the post-Lab 9 completion sprint.

5.3 Concrete Roadmap to Complete the Project (Final Presentation ~31 Oct 2026)

With Lab 9 concluding the formal laboratory sessions, our team will utilize the remaining 2.5 weeks to transition Dash from a validated lab build into a polished final project showcase for the **31 October 2026** evaluation:

Timeline Sprint	Core Engineering & Design Deliverables	Tangible Milestone Deliverable
Sprint 1 **(15–20 Oct 2026)**	• Integrate supporting AI Natural Language Assistant (slot-filling intent parser for spoken/typed queries). • Refine mobile touch targets to 48px minimum across all screen viewports. • Harden offline fallback caching for the RailKit API adapter.	**Build v3.1 Feature Complete** (AI Assistant & Mobile Polish)
Sprint 2 **(21–26 Oct 2026)**	• Deploy production staging instance to Vercel with automated CI/CD. • Conduct secondary 10-user usability stress test to verify AI chat interaction flows. • Achieve 100/100 Google Lighthouse Performance & Accessibility score.	**Public Staging URL & Stress Test Log** (Zero-Defect Production Candidate)

Timeline Sprint	Core Engineering & Design Deliverables	Tangible Milestone Deliverable
Sprint 3 **(27–31 Oct 2026)**	• Package high-resolution demonstration video walkthrough showcasing live RailKit integration. • Author the comprehensive Final Project Dossier & Slide Deck. • Rehearse interactive viva defense and live examiner demonstration script.	**Final Project Presentation & Defense** (31 Oct 2026 Showcase)

6. Viva Defense Guide (Evaluation Plan: 2 Marks)

This defense guide prepares team members **Harshit** and **Uttam Kumar** to answer rigorous examiner questions covering the complete design journey (Labs 4–9) and foundational evaluation theories from *Lab 3 (Slides 8–10)*:

Q1: How did the three evaluation types from Lab 3 (Slide 8–10) govern your Lab 9 methodology?

Defense Answer: *"In Lab 3, we learned that complete interface evaluation requires triangulating three distinct paradigms: (1) Usability Inspection (expert heuristic evaluation) to catch structural layout flaws without user overhead; (2) User-Based Evaluation (empirical think-aloud and SUS) to observe authentic behavioral friction and cognitive load; and (3) Accessibility Evaluation (Lighthouse/WAVE/WCAG) to ensure universal accessibility. In Lab 9, we utilized all three: we re-audited Nielsen's 10 heuristics, tested 5 brand-new users across our 2 Lab 5 tasks, and conducted automated WCAG AA contrast/keyboard checks."*

Q2: Why did you recruit 5 completely new participants rather than re-testing your Lab 7 users?

Defense Answer: *"Testing the same participants would introduce severe 'learnability bias' (or carry-over effects). Participants from Lab 7 already possessed mental models of Dash's structure, which would falsely inflate completion speeds and mask remaining usability defects. Recruiting 5 fresh participants (N01–N05) adhered to Nielsen's classic 5-user testing rule while guaranteeing an unbiased, ecologically valid measurement of the improved build."*

Q3: Pick one major defect from Lab 7 and trace its full trajectory through Lab 8 and Lab 9.

Defense Answer: *"In Lab 7, Issue #3 (Delhi Terminal Ambiguity) caused Participant P03 to hesitate for 20s because the system forced choosing between NDLS, DLI, and DEC with no geographic context. In Lab 8, we redesigned the search architecture to default to a 'Delhi — All Stations' meta-cluster and introduced an intelligent Station Intelligence panel comparing arrival speeds. In Lab 9, new user N01 completed station selection in 2.8s, praising the instant clarity that Delhi Cantt was fastest for Gurgaon commuters. Post-validation severity dropped from 3 (Major) to 0 (Resolved)."*

Q4: How did you fix the Catastrophic issue regarding Live Availability trust (Issue #5)?

Defense Answer: *"In Lab 7, users doubted whether seat counts were real or simulated. In Lab 8/9, we integrated a real-time server-side API adapter powered by `railkit@6.0.1` connecting directly to live IRCTC availability. We introduced a 3-tier cascade: primary RailKit query, secondary public proxy fallback, and transparent baseline. Whenever live data is fetched, a high-visibility green 'LIVE' badge with a sync timestamp is rendered. If upstream railway servers timeout, Dash explicitly displays a 'DEMO FALLBACK' badge rather than masquerading fake numbers as live."*

Q5: How does your Group Constraint Engine embody Nielsen's 'Flexibility and Efficiency of Use' heuristic?

Defense Answer: *"Traditional IRCTC forms force users to manually navigate multiple screens to request lower berths for elderly travelers. Our Group Constraint Engine automatically detects passenger age (≥ 60) and applies intelligent constraints (lower-berth priority and adult-child proximity) inline. Furthermore, we solved the classic multi-leg limitation by allowing per-leg berth preference overrides directly during planning without wiping the master list."*

Verification & Evaluator Sign-Off

We hereby certify that this Lab 9 Validation Summary Report reflects authentic empirical testing, rigorous heuristic re-inspection, and automated accessibility auditing conducted for Course AI511 at the Indian Institute of Technology Ropar.

Lead Evaluator: Harshit
Course Role: AI511 Project Lead / Evaluator
Signature: *Harshit (Verified)*
Date: 14 October 2026

Co-Evaluator: Uttam Kumar
Course Role: AI511 Co-Evaluator / Teammate
Signature: *Uttam Kumar (Verified)*
Date: 14 October 2026